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Carpenter Custom 455® Stainless Steel, Annealed, 25 mm Bar

Categories: [Metal](#); [Ferrous Metal](#); [Stainless Steel](#); [T 400 Series Stainless Steel](#)

Material Data provided by Carpenter Technology Corporation.

Notes: Recognizing the need for high-strength alloys with good corrosion resistance to atmospheric environments, the Carpenter Research Laboratory developed Custom 455® stainless, a martensitic age-hardenable stainless steel. This alloy is relatively soft and formable in the annealed condition. A single-step aging treatment develops exceptionally high yield strength with good ductility and toughness. This stainless can be machined in the annealed condition, and welded in much the same manner as other precipitation hardenable stainless steels. Because of its low work-hardening rate, it can be extensively cold formed. The dimensional change during hardening is only about -0.001 in/in, which permits close-tolerance finish machining in the annealed state. Custom 455 stainless represents a significant advancement in the area of precipitation hardening stainless steels. It should be considered where simplicity of heat treatment, ease of fabrication, high strength and corrosion resistance are required in combination.

Because of the unique combination of high strength and corrosion resistance of Custom 455 stainless there are few other alloys available for consideration. Carpenter PH13-8 Mo can be considered where good transverse toughness and ductility are necessary in large sections.

Custom 455® is a registered trademark of Carpenter Technology Corporation.

Key Words: UNS S45500

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Physical Properties	Original Value	Comments
Density	7.76 g/cc	
Mechanical Properties	Original Value	Comments
Hardness, Brinell	297	Estimated from Rockwell C for Brinell test with 3000 kg load, 10 mm ball.
Hardness, Knoop	322	Estimated from Rockwell C
Hardness, Rockwell C	31	
Hardness, Vickers	311	Estimated from Rock
Tensile Strength, Ultimate	1000 MPa	C

Tensile Strength, Yield	713 MPa @Strain 0.2 %	
Elongation at Break	14 %	In 4D
Reduction of Area	60 %	
Modulus of Elasticity	200 GPa	
Poissons Ratio	0.30 @Temperature 23.0 °C	
Shear Modulus	76.9 GPa @Temperature 23.0 °C	
Charpy Impact	95.0 J	V-notch
Electrical Properties	Original Value	Comments
Electrical Resistivity	0.0000906 ohm-cm	
Thermal Properties	Original Value	Comments
CTE, linear	10.6 µm/m-°C @Temperature 22.0 - 93.0 °C	
	11.2 µm/m-°C @Temperature 22.0 - 260 °C	
	12.0 µm/m-°C @Temperature 22.0 - 482 °C	
Thermal Conductivity	18.0 W/m-K @Temperature 100 °C	
	21.3 W/m-K @Temperature 300 °C	
	24.7 W/m-K @Temperature 500 °C	
Component Elements Properties	Original Value	Comments
Carbon, C	<= 0.050 %	
Chromium, Cr	11 - 12.5 %	
Copper, Cu	1.5 - 2.5 %	
Iron, Fe	75 %	as remainder
Manganese, Mn	<= 0.50 %	
Molybdenum, Mo	<= 0.50 %	
Nb + Ta	0.10 - 0.50 %	
Nickel, Ni	7.5 - 9.5 %	
Niobium, Nb (Columbium, Cb)	<= 0.50 %	
Phosphorus, P	<= 0.040 %	
Silicon, Si	<= 0.50 %	
Sulfur, S	<= 0.030 %	
Tantalum, Ta	<= 0.50 %	
Titanium, Ti	0.80 - 1.4 %	

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